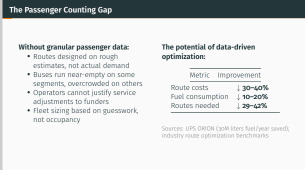
A Smarter Public Transport

The **inMotion** project aims to bring real-time intelligence to public transport operations.
One of its core goals, understand **how passengers move** through the network: who boards, who alights, and where.
Today we present a component of that vision: passenger movement classification using only Wi-Fi signals.

Smart bus concept

- Welcome everyone. This work is part of the inMotion project, funded by CENTRO 2030.
- The project's mission is to bring intelligence to public transport operations.
- Today I'll focus on one specific challenge: how do we know how many people board and alight at each stop, without installing expensive hardware on every bus?
- The key insight: we use Wi-Fi signals that are already there.



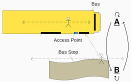
- Accurate passenger counting is fundamental for public transport operations. Without it, operators are flying blind.
- Traditional systems, infrared beams, pressure mats, cameras, work reasonably well in labs. In the real world, with crowds, occlusion, and dirt, their accuracy drops dramatically — 98% claimed becomes 53% in practice, as Pronello and Ruiz documented.
- But beyond the technology problem, there is a financial case. Why does accurate counting matter? Because data is what enables route optimization.
- Look at the logistics industry: UPS saved 30 million liters of fuel per year just by optimizing routes with their ORION system. Shein and TikTok Shop cut delivery costs by 30–40% using AI route planning.
- The same principle applies to public transport. If you know exactly where passengers board and alight, you can right-size vehicles, adjust frequencies, and eliminate empty runs. That is the economic motivation behind this work.
- Wi-Fi is already on many buses. The infrastructure is there. We just need to extract the

Four Movements to Classify

We want to answer a simple question for every passenger interacting with the bus door:
Did this person board, alight, or stay put?

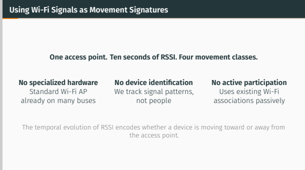
Label	Meaning	Description
AA	$A \to A$	Remaining inside the bus
BB	$B \to B$	Remaining at the bus stop
BA	$B \to A$	**Boarding** the bus
AB	$A \to B$	**Alighting** the bus

Zone A = bus interior (near access point)
Zone B = bus stop (outside)



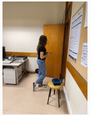
Two-zone environment: door separates A from B

- Let me formalize the problem. We have a bus door, and an access point nearby.
- Zone A is inside the bus. Zone B is the outside, the bus stop area.
- Any person near the door is in one of four states: inside and staying inside (AA), outside and staying outside (BB), moving from outside to inside, boarding (BA), or moving from inside to outside, alighting (AB).
- This is the fundamental classification we need to solve.
- The notation uses arrows: B→A means boarding, A→B means alighting, and the static classes are AA and BB.



- The core idea is simple. If you hold a Wi-Fi device and walk toward an access point, the signal gets stronger. Walk away, it gets weaker. Stay still, it stays flat.
- We capture this over a 10-second window at 1 sample per second, 10 RSSI values.
- That is all we need. One AP at the door. No cameras. No pressure mats. No special hardware.
- And critically: we do not identify devices. We just need to know that a signal exists and how it changes. The privacy implication is fundamental.
- The question is whether machine learning can learn these temporal patterns reliably.

Experimental Setup

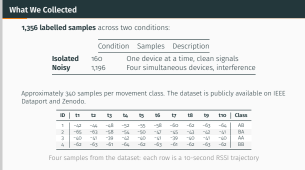


Controlled indoor environment at IT Aveiro

Physical setup:
- Zone A: closed room (bus interior), AP at doorway
- Zone B: corridor (bus stop), wall attenuation between zones

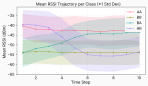
Collection:
- 10 s windows at 1 Hz → 10 RSSI values per sample
- 4 smartphones, 3 brands
- Scripted movements with ground truth labels
- MAC-based device isolation for clean trajectories

- We built a controlled environment at our institute. A closed room represents the bus interior, and the corridor outside represents the bus stop.
- The access point sits right at the doorway, exactly where you would mount it on a real bus.
- We used four different smartphone models: Samsung Galaxy S20, Samsung Galaxy S23, POCO X7 Pro, and Xiaomi Redmi 4. Three brands, spanning Android 6 to Android 14, different Wi-Fi chipsets. This diversity is essential for generalization.
- Each trial is exactly 10 seconds long, with RSSI sampled once per second, so 10 values per observation window.
- In the noisy scenario, four devices operated simultaneously in paired movements — some boarding while others stayed static, creating realistic co-channel interference.
- Because we controlled the environment, we knew the ground truth for every trial, each 10-second window was labeled as AA, BB, AB, or BA.



- Our dataset has 1,356 samples, each one a 10-second window with 10 RSSI values and a class label.
- We collected under two conditions: isolated (single device, clean signals) and noisy (four devices transmitting simultaneously).
- The noisy scenario is much closer to reality: in a real bus stop you have multiple people, some boarding, some waiting, some just passing by. These additional devices create overlapping RSSI patterns, and the classifier must cut through that noise.
- The isolated subset is small, only 160 samples, but it gives us an upper bound on what is possible without interference.
- The dataset is publicly available on IEEE Dataport and Zenodo. We want other researchers to build on this work.
- Look at this sample: rows 1 and 2 show alighting (decreasing RSSI) and boarding (increasing RSSI). These temporal patterns, not the absolute values, are what the classifier learns.

Temporal RSSI Trajectories



Mean RSSI trajectory per class over 10 seconds (± 1 std)

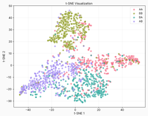
Clear temporal signatures emerge:

- **Boarding (BA):** steady RSSI increase as the device approaches the AP
- **Alighting (AB):** pronounced RSSI decrease as the device moves away
- **Static states (AA, BB):** stable signal at distinct levels: AA higher (near AP), BB lower (outside)

These contrasting trends are the foundation of the classification.

- Here we see why the problem is solvable. Each movement class has a distinctive RSSI signature over time.
- Boarding, the green line, shows signal strength climbing as the person walks toward the access point.
- Alighting, the red line, shows signal dropping as they walk away through the door.
- The two static classes sit at different levels: AA is higher because the device is inside near the AP; BB is lower because it is outside, behind a wall.
- There is overlap in standard deviations, this is not a trivial problem. But the mean trends are distinct enough that machine learning can separate them.
- Remember: each of these lines is just 10 RSSI values. That is our entire feature space.

t-SNE Visualization

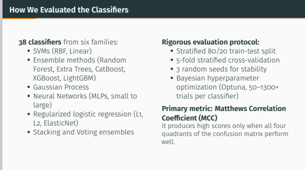


t-SNE projection of the 10-dimensional RSSI feature space

Classes form well-defined clusters confirming that the collected samples carry structure for classification.

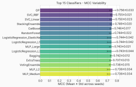
Overlap between AA and BA expected: both involve strong RSSI near the AP, but differ in temporal pattern.

- t-SNE projects our 10-dimensional data into 2D so we can see the structure.
- The four classes form distinct clusters. They are not perfectly separated, but they are clearly structured.
- The overlap between AA and BA makes physical sense: both involve a device near the access point with high RSSI. The difference is whether the signal is stable (AA) or increasing (BA).
- This overlap is what makes the problem interesting, and where machine learning adds value over simple thresholding.
- The important message: yes, 10 RSSI values are enough. The data has structure.



- We did not just try one or two classifiers. We evaluated 38 different algorithms across six families.
- Every classifier underwent Bayesian hyperparameter optimization with Optuna, with up to 1,300 trials per model. This ensures a fair comparison.
- We used three random seeds and report mean and standard deviation. Reproducibility matters.
- Now, the most important metric. We chose MCC, Matthews Correlation Coefficient, as our primary metric. Why? Because accuracy alone is misleading for multi-class problems. MCC only gives high scores when the model performs well across all four confusion matrix quadrants, true positives, true negatives, false positives, and false negatives for every class.
- MCC is the standard in biomedical machine learning and increasingly in other fields. It is a single number you can trust.

MCC Comparison



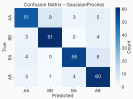
MCC variability across three random seeds (combined dataset)

Classifier	Combined	Isolated	Noisy
GP	**0.756**	0.414	0.755
SVC (RBF)	0.755	0.825	0.754
CatBoost	0.746	0.782	**0.770**
KNN (k=5)	0.692	**0.907**	0.704

Mean MCC ± std across 3 seeds

- This table shows our headline results across the three scenarios.
- Gaussian Process achieved the best MCC on the combined dataset at 0.756.
- Look at the isolated scenario: KNN hits 0.907 MCC, near-perfect classification when there is no interference. This tells us the signal patterns themselves are highly discriminative.
- But Gaussian Process on isolated drops to 0.414. Why? The isolated set has only 160 samples, too few for the GP kernel to estimate reliably. This is a sample size limitation, not a model failure.
- The noisy scenario, closest to reality, shows CatBoost at 0.770. That is our most realistic performance estimate.
- The noisy results are consistent with the combined results because noisy data makes up 88% of the dataset.

Gaussian Process on Combined Dataset



Normalized confusion matrix, Gaussian Process

Class	Precision	Recall	F1
AA (Inside)	0.78	0.73	0.75
BB (Stop)	0.87	0.89	0.88
BA (Boarding)	0.83	0.77	0.80
AB (Alighting)	0.78	0.88	0.83
Weighted Avg	**0.82**	**0.82**	**0.82**

Strong diagonal dominance. Errors are concentrated between spatially adjacent classes: AA/BA and AB/BB.

- The confusion matrix shows where we succeed and where we struggle.
- Diagonal dominance is strong. Most predictions are correct.
- The main confusions are between AA and BA, both near the access point, both with strong RSSI, and between AB and BB, both with weaker signals.
- Notice that BB (waiting at the stop) is the easiest class: 0.88 F1. The physical barrier between zones creates a clear signal separation.
- AA (inside the bus) is the hardest: 0.75 F1. When a device is already near the AP, the temporal signal of boarding versus staying is subtle.
- This tells us where future work should focus: distinguishing static from transitional states when the RSSI magnitude is similar.

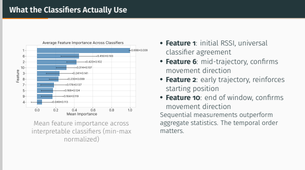
The Effect of Interference

Scenario	Best Classifier	MCC	AA F1	BB F1	BA F1	AB F1
Isolated	KNN (k=5)	0.907	0.87	0.96	0.91	0.96
Noisy	CatBoost	0.770	0.79	0.87	0.82	0.83
Combined	Gaussian Proc.	0.756	0.75	0.88	0.80	0.83

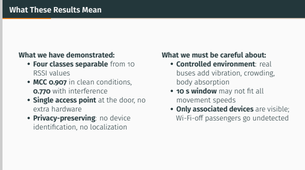
Clean signals: near-perfect
F1 scores above 0.86 for every class.
Simple classifiers (KNN) perform best.
Upper bound on what RSSI can achieve.

Realistic conditions: robust
Per-class F1 above 0.77 with interference.
Ensemble methods handle noise best.

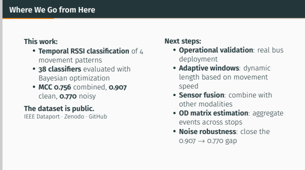
- This slide makes the clean-versus-noisy comparison explicit.
- Under clean, isolated conditions, we get near-perfect classification. KNN with k=5 reaches 0.907 MCC. BB and AB have F1 scores of 0.96. This shows the promise of the approach.
- Under noisy, realistic conditions, performance drops but remains solid. CatBoost achieves 0.770 MCC, with per-class F1 above 0.77. Ensemble methods handle interference better than simpler classifiers.
- The gap between 0.907 and 0.770 is the cost of interference. Closing this gap, making the system robust to noise, is our main focus going forward.
- But even 0.770 MCC is operationally useful as a complementary technology. It does not need to replace APC systems; it augments them.
- Gradient boosting methods like CatBoost and XGBoost showed particular robustness to signal interference, making them our recommended choice for deployment.



- Feature importance reveals that the first RSSI measurement is overwhelmingly the most important, all classifiers agree on this (standard deviation 0.009).
- The first three samples essentially tell the classifier where the device starts. Feature 6, at the middle of the window, captures whether the device is moving and in which direction.
- The later features (7–10) matter less because by mid-window, the trajectory is largely determined.
- This has a practical implication: we might be able to shorten the observation window and still maintain performance. That matters for faster boarding scenarios.
- The key insight: temporal order matters. Sequential RSSI values carry information that aggregate statistics like mean and variance lose.



- Let me be clear about what we have and have not shown.
- We have demonstrated that the approach works in a controlled environment. MCC of 0.907 in clean conditions proves the signal patterns are discriminative. MCC of 0.770 in noisy conditions shows the approach is robust enough to be useful.
- But I want to be transparent about limitations. This is a lab experiment. On a real bus, you have vehicle vibration, metal surfaces reflecting signals, crowds of people absorbing RF, and passengers moving at different speeds. All of these degrade RSSI separability.
- Also, our system only sees devices that are associated with the access point. Passengers with Wi-Fi off, using mobile data exclusively, or whose devices do not auto-connect are invisible. Modern MAC address randomization during probe requests does not affect us since we rely on Wi-Fi association, but first-time passengers who never connect will not be counted.
- The isolated dataset is small. We report those numbers because they are informative, but do not overinterpret them.



- To wrap up: we proposed a new way to classify passenger movements using only Wi-Fi RSSI. No cameras, no pressure mats, no specialized hardware.
- The results are promising, not definitive. We showed that the approach works in controlled conditions. The next step is to take it to a real bus.
- We are particularly excited about adaptive observation windows: can the system decide how long to observe based on what it sees?
- Sensor fusion is another key direction: combining RSSI with other low-cost modalities could make the system robust enough for production deployment.
- OD matrix estimation is the ultimate goal: aggregate individual boarding and alighting events across stops to recover full origin-destination flows for the network.
- The dataset is public. We encourage the community to build on this work, try new classifiers, new features, new deployment scenarios.
- Finally, we want to close the gap between clean (0.907) and noisy (0.770). That 0.137 MCC difference is where the real research lives.

└─Acknowledgements

Acknowledgements

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Thank you. Questions?

Website: atnog.github.io/inMotion · Dataset: IEEE Dataport (10.21227/55nm-0r91) · Code: github.com/ATNoG/inMotion

- Before I take questions, I want to acknowledge the funding that made this possible. This work is part of the inMotion project, funded by European structural funds through CENTRO 2030.
- The dataset and code are publicly available, the links are on the slide.
- I am happy to take your questions now.